

Module-2

1. Write the differences between supervised and unsupervised learning.

<https://anikchandml.hashnode.dev/2-supervised-vs-unsupervised-learning>

2. Write down the differences between classification and clustering.

Classification vs Clustering

Feature	Classification	Clustering
Type of Learning	Supervised Learning	Unsupervised Learning
Data Labels	Requires labeled data	Uses unlabeled data
Goal	Predict the class/category of new data	Group similar data points together
Output	Predefined classes	Clusters formed automatically
Training Data	Input + correct output available	Only input data available
Example	Email → Spam or Not Spam	Group customers based on purchasing behavior
Algorithms	Decision Tree, Random Forest, SVM, Naive Bayes, Logistic Regression	K-Means, Hierarchical Clustering, DBSCAN
Accuracy Measurement	Accuracy, Precision, Recall, F1-score	Silhouette Score, Davies-Bouldin Index

3. What do you mean by clustering?

4. Write note on the different types of clustering.

<https://anikchandml.hashnode.dev/12-k-medoids-clustering>

<https://anikchandml.hashnode.dev/11-k-means-clustering>

k-means algorithm:

1. Explain briefly the working principles of k-means algorithm.
2. What are the differences between K-means clustering and K-nearest neighbor (KNN).
3. Numerical question.
4. How to decide the optimal number of K in the K means Algorithm?
5. What are the advantages and disadvantages of the K means Algorithm?
6. Write the working principle of the K-means algorithm.
7. Use k-means algorithm to evaluate clusters in given data.
8. How does K-Medoids differ from K-Means in choosing cluster centers?
9. How to determine K using the ELbow method?
10. Differentiate between K-means clustering and K-nearest neighbor (KNN).

<https://anikchandml.hashnode.dev/11-k-means-clustering>

Difference Between K-Means and K-Medoids

Feature	K-Means	K-Medoids
Cluster Center	Mean (average) of points	Actual data point (medoid)
Center Value	May not exist in dataset	Must be one of the dataset points
Sensitivity to Outliers	High	Low
Distance Measure	Usually Euclidean distance	Any distance metric can be used
Computation Speed	Faster	Slower
Accuracy with Noisy Data	Lower	Higher
Memory Requirement	Less	More
Suitable For	Large datasets with few outliers	Datasets containing noise and outliers

Feature	K-Means	KNN	
Type	Clustering Algorithm	Classification/Regression Algorithm	
Learning	Unsupervised Learning	Supervised Learning	
Data Labels Required?	No	Yes	
Purpose	Groups similar data points into clusters	Predicts the class/value of a new data point	
Training Data	Unlabeled data	Labeled data	
Output	Clusters	Class label or predicted value	
Meaning of K	Number of clusters	Number of nearest neighbors	
Working Principle	Uses centroids and minimizes within-cluster distance	Uses majority voting (classification) or averaging (regression) among nearest neighbors	
Example	Group customers into different segments	Predict whether an email is Spam or Not Spam	
Model Building	Creates cluster centroids during training	No explicit model; stores training data	

PCA:

- 1.Explain the term “Curse of Dimensionality”.
- 2.What are the ways of dealing with sparse features?
- 3.What are the common problems with sparse dataset in machine learning?
- 4.Why is machine learning difficult with sparse features?
- 5.What is difference between sparse data and missing data?
- 6.What is principal component analysis?
- 7.What does a PCA do?
- 8.What are the steps involved in PCA algorithm.
- 9.What are the advantages and disadvantages of PCA algorithm?
10. How does PCA transform correlated features into uncorrelated ones?
- 11.What are the benefits and limitations of the PCA algorithm?
- 12.Why is it necessary to standardize data before applying PCA?
- 13.Describe the role of eigenvalues and eigenvectors in PCA.

<https://anikchandml.hashnode.dev/13-pca>

1. Short note on Matrix Factorizⁿ.

Matrix Factorization is a technique used to decompose a large matrix into two or more smaller matrices. It helps discover hidden patterns and relationships in data while reducing dimensionality.

Suppose a matrix R represents user ratings for movies. Matrix Factorization decomposes it into:

$$R \approx P \times Q$$

where:

- R = Original matrix
- P = User feature matrix
- Q = Item (movie) feature matrix

The product of these smaller matrices approximates the original matrix and can be used to predict missing values.

Advantages

- Reduces dimensionality
- Handles sparse data effectively
- Discovers latent (hidden) features
- Improves recommendation accuracy

Example

If a user has not rated a movie, Matrix Factorization can estimate the rating based on learned user and movie characteristics.

2. Usage of Matrix Factorization

Matrix Factorization is widely used in:

1. Recommendation Systems

- Netflix movie recommendations
- Amazon product recommendations
- Spotify music recommendations
- YouTube video suggestions

2. Collaborative Filtering

Predicts user preferences based on past behavior and ratings.

3. Dimensionality Reduction

Reduces large datasets into smaller representations while preserving important information.

4. Image Processing

Used for image compression and feature extraction.

5. Text Mining and NLP

Helps discover hidden topics in documents and large text datasets.

6. Data Compression

Represents large matrices using smaller matrices to save storage and computation.

Key Idea

Matrix Factorization breaks a large matrix into smaller matrices to uncover hidden patterns and make predictions, especially in recommendation systems.